



Data Analytics

Portfolio of Miriyam Revathi

Professional Summary

My name is **Miriyam Revathi**, and I am currently pursuing my **3-2** of course CSE(AI&ML) at **Narasaraopeta Engineering College**. I Was highly motivated in **Data Analytics, Python programming, SQL, and Machine Learning**, which have strengthened my understanding of data-driven problem solving and analytical thinking.

I worked on multiple analytics projects across domains such as **banking, social media, operations, hiring analytics, movie analysis, and call center optimization**. These projects involved working with raw and structured data, identifying data quality issues, applying analytical techniques, and presenting insights in a clear and professional manner.

I have developed strong practical skills in **Data Analytics using Excel ,SQL,Power BI and Tableau** as well as **Python and Machine Learning**. I have worked on multiple real-world projects involving **data cleaning, exploratory data analysis (EDA), data visualization, and business insight generation**. I am also actively building my Machine Learning foundation, focusing on **data preprocessing, feature engineering, supervised learning algorithms, and model evaluation**, and applying these concepts through hands-on practice and ongoing projects.

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MODULE 1 PROJECT :Data Analytics Process

Objective:

Understand and apply the complete data analytics lifecycle.

Topics Covered:

- Business Understanding
- Data Collection
- Data Cleaning
- Data Analysis
- Visualization
- Communication of Results

Approach:

I followed a structured analytics framework to solve a recommendation-based business problem and 5 why's rule. The project emphasized understanding user behavior, cleaning irrelevant data, extracting insights, and presenting recommendations clearly.

Tools Used:

MS Excel

Power BI

Tableau

Outcome:

This project built my foundation in thinking like a data analyst and understanding how data-driven decisions are made.

Module 2 – Hiring Process Analytics

Project Overview

Analyzed a company's hiring dataset to understand **hiring trends, salary structure, and workforce distribution**, supporting HR and management decision-making.

Methodology

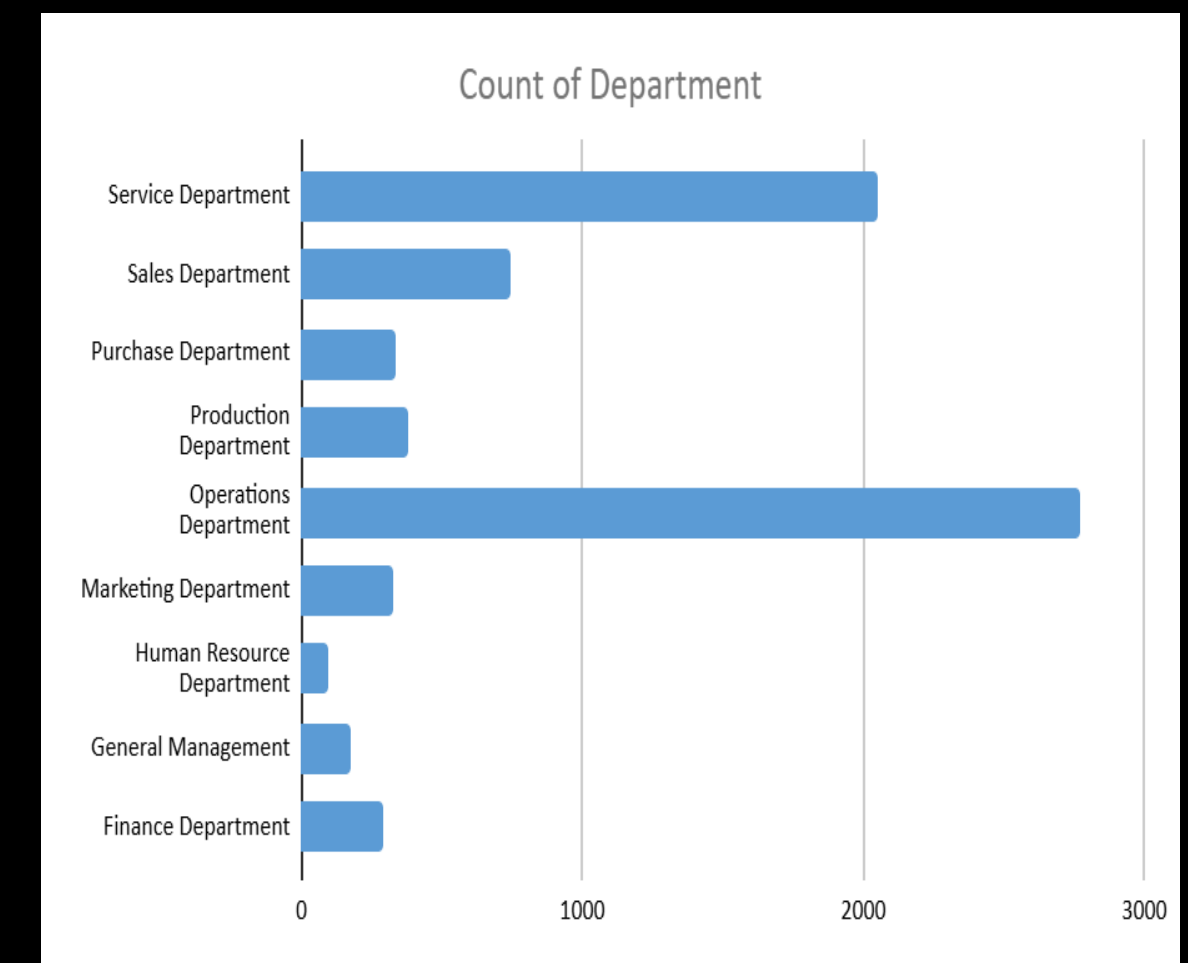
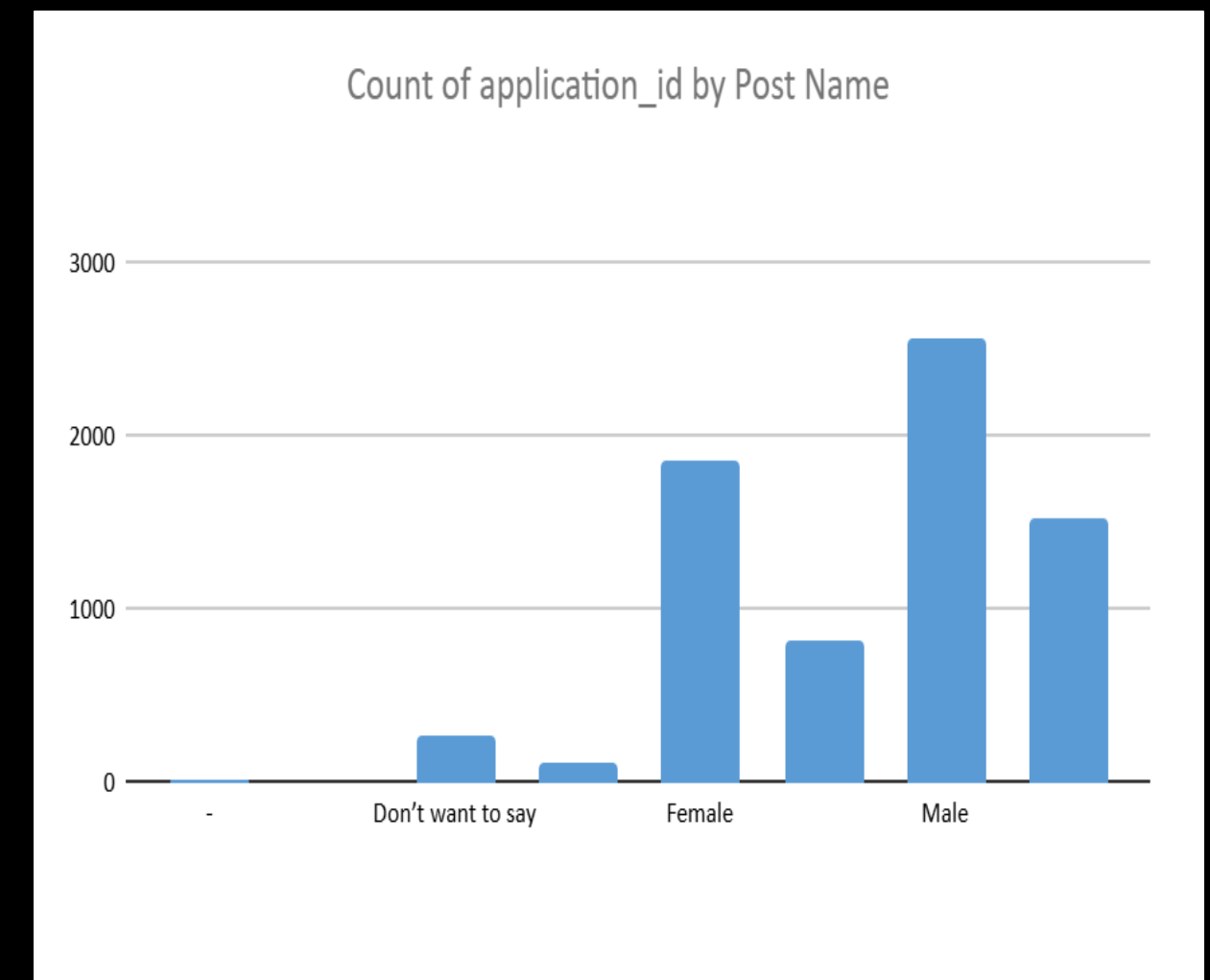
Explored a **7-column hiring dataset**, cleaned missing values in event_name and post_name (15 rows), and used Excel formulas and charts to answer business questions related to hiring and compensation.

Key Analysis Performed

- **Hiring Analysis:** Gender-wise hiring distribution — **1,856 females, 2,563 males, 268 unspecified**.
- **Salary Analysis:** Calculated average salary (**~49,933**) and identified outliers using MIN (**100**) and MAX (**400,000**).
- **Salary Distribution:** Created salary bands using $(max - min)/9$.
- **Department & Role Analysis:** Visualized headcount by department and position tier; **Operations** had the highest workforce, **HR** the lowest.
- **Tools Used**
Microsoft Excel (**2023**) – statistical functions, charts; **PowerPoint** – reporting.

Results & Insights

Identified gender imbalance, extreme salary outliers, and uneven department



Module 3 – IMDB Movie Analysis

Overview

analyzed an IMDB movie dataset to study how **genre, duration, language, director, and budget** impact **ratings and financial performance**.

Data preparation

cleaned **5,000+ records** by removing duplicates and handling missing values to ensure reliable statistical analysis.

Key analysis

- Identified **drama** as the most common genre and analyzed imdb scores by genre.
- Studied the relationship between **movie duration and ratings** using trendlines.
- Found **english** as the dominant language and compared ratings across languages.
- Ranked directors using percentile analysis; many top directors had limited movies.
- Analyzed **budget vs gross**, calculated profit and profit margins; **avatar** showed the highest profit and gross with low budget–gross correlation (**~0.10**).

Insights

higher budgets often improve ratings, but **strong storytelling and direction** allow some low-budget films to succeed. Movie success depends on A mix of **creative quality, audience appeal, and smart budgeting**, with moderate-budget films sometimes yielding the best returns.

Module 4 – Bank Loan Case Study

Overview

Analyzed bank loan application data to identify **credit risk factors** and support **better loan approval decisions** using exploratory data analysis (EDA).

Methodology

Followed a structured EDA pipeline including **data cleaning, univariate and bivariate analysis, outlier detection, class imbalance checks, correlation analysis, and visualization** to evaluate customer and loan characteristics.

Key Analysis

Handled missing values by categorizing columns into **numeric, categorical, and binary**, using **median, mode, and majority-class imputation** respectively.

- Detected **outliers** using statistical measures (min, max, quartiles, bounds).
- Analyzed **class imbalance** between repayers and defaulters.
- Performed univariate, segmented, and bivariate analysis using pivot tables and charts.
- Used **correlation analysis and heatmaps** to identify key default drivers such as **income, credit amount, annuity (EMI), and loan-to-income ratio**.

Tools Used

Microsoft **Excel 2023, Power Query**, Excel functions and **VBA** for automation.

Insights & Results

The analysis revealed strong **class imbalance**, wide variation in income and credit amounts, and higher default risk among **younger, lower-income applicants with more dependents**. The **loan-to-income ratio** emerged as a critical default indicator. The final Excel model enables quick risk assessment, data quality checks, and supports **real-world banking credit decisions** while strengthening Excel-based risk analytics skills.

Module 5 – Instagram User Analytics

Project Overview

This project focuses on analyzing an Instagram-like dataset to understand **user engagement and interaction patterns**. As a Data Analyst, I worked with data related to **users, posts, likes, comments, follows, and hashtags** to extract actionable insights that can support **product decisions, marketing strategies**.

Methodology

A structured analytics approach was followed. First, the business goals were defined based on engagement and growth metrics. Then, the relational tables (users, photos, likes, comments, follows, tags, photo_tags) were explored to understand their relationships. Data cleaning was handled by selecting only relevant records through proper joins. Finally, **SQL queries** were written to aggregate data, identify patterns, and answer business questions using **MySQL 8.0 and MySQL Workbench**.

Marketing Analysis:

- Identified the **top 5 oldest users** to reward loyal customers.
- Detected **inactive users** using LEFT JOINs (users with no posts).
- Found the **most liked photo** to declare a contest winner.
- Ranked **top hashtags** to support hashtag research.
- Identified the **day with maximum user registrations** for ad campaign planning.

Results and Insights

The analysis revealed that long-time users are not always active, and many users have never posted, indicating opportunities for re-engagement. Certain posts achieved viral reach with very high like counts, while popular hashtags consistently drove visibility. User sign-ups peaked on specific days, helping optimize marketing timing. Overall posting frequency was lower than expected, and the project successfully identified **active, inactive, and suspicious accounts**, strengthening real-world SQL and data analysis skills.

Module 6 – Operation Analytics & Metric Spike Investigation

Overview

Analyzed operational and user activity data to track **performance, engagement, and metric stability**.

Key Analysis

- Calculated jobs reviewed over time and **7-day rolling throughput**.
- Analyzed **language share**, weekly user engagement, growth, and retention.
- Studied **device-wise and email engagement** patterns.
- Verified data quality (no duplicate records).

Tools Used

MySQL 8.0, MySQL Workbench, SQL, Excel (data formatting).

Insights

Observed **stable throughput**, varying user engagement trends, dominant language usage, and clear device preferences.

The analysis helped identify engagement drops and areas for operational improvement

Module 7 – ABC Call Volume Trend Analysis

Overview

Analyzed **23 days of inbound call center data** to understand call volume trends, average call duration, and design an optimized **manpower plan** to reduce call abandonment and improve customer experience.

Key Analysis

Analyzed **hour-wise call volume trends** to identify peak and off-peak periods.

Calculated **average call duration (~3.3 minutes)** to estimate agent handling capacity.

Designed **day-shift (9 AM–9 PM) manpower planning** to ensure 90% calls answered.

Estimated **night-shift (9 PM–9 AM) staffing** based on reduced call volume assumptions.

Identified data quality issue where **NULL Agent IDs (till Jan 13)** caused automatic call abandonment.

Tools Used

Microsoft Excel (data cleaning, pivot tables, formulas, charts), **PowerPoint** (reporting).

Insights & Results

Call volume varied significantly across the day, with peaks in late morning and evening. High abandonment (~30%) was mainly due to insufficient staffing during peak hours and no agent availability at night. The optimized manpower model reduced abandonment to **~10%**, improved service levels, and demonstrated how **data-driven workforce planning** directly impacts call center efficiency and customer experience.

Module 8 –Analyzing the Impact of Car Features on Price & Profitability

overview

Analyzed car market data to understand how **engine power, fuel efficiency, body style, transmission, and brand** influence **car price and profitability**.

Key Analysis

Cleaned a large dataset (**11,000+ rows**) by removing duplicates, handling missing values, and standardizing categories.

Used **pivot tables and visualizations** to analyze price variation across brands, body styles, and market categories.

Studied relationships between **horsepower, cylinders, MPG, and MSRP** using charts and regression analysis.

Identified key features driving higher prices and popularity.

Tools Used

Microsoft Excel, Pivot Tables, Charts, Data Analysis ToolPak (regression), **Interactive Dashboard**.

Insights

Cars with higher **engine power and more cylinders** generally have higher prices, while luxury brands and larger body styles command higher MSRPs. Fuel efficiency decreases with more cylinders, and certain market categories show higher popularity. The final **Excel dashboard** enables quick exploration of price, performance, and feature relationships to support **pricing, product, and marketing decisions**

Overview

Built an **interactive Power BI dashboard** to analyze Amazon store sales, profit, orders, and customer behavior across regions, categories, and time.

Key Analysis

Tracked KPIs: **Sales (2M), Profit (175K), Orders (5901), Products (1755)**

Analyzed sales by **segment, category, sub-category**

Studied **monthly and quarterly sales trends**

Visualized **state-wise sales** using maps

Evaluated **payment modes, shipping methods, and return status**

Tools Used

Power BI Desktop, Power Query, DAX, interactive charts and slicers

Insights

Consumer segment and Office Supplies drive highest sales, Q4 shows peak performance, standard shipping dominates, and return rates are low.

AMAZON STORE SALES DASHBOARD

Central

East

South

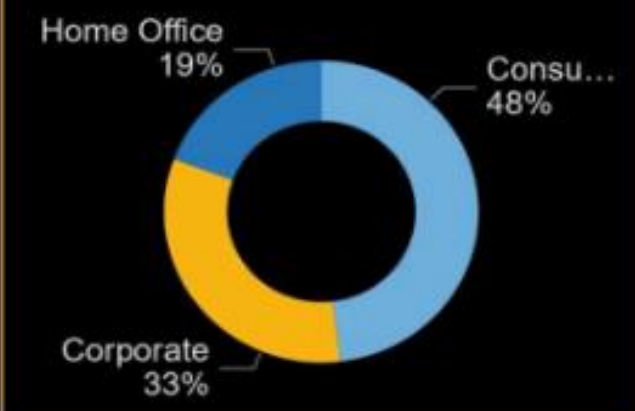
West

Choose State

All



Sales by Segment



Sales

2M

Profit

175K

Orders

5901

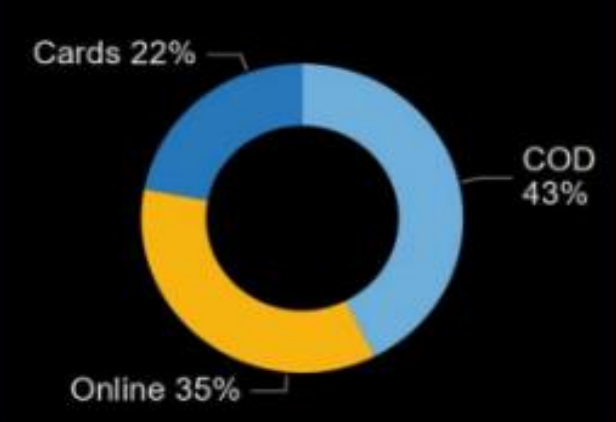
Total Products

1755

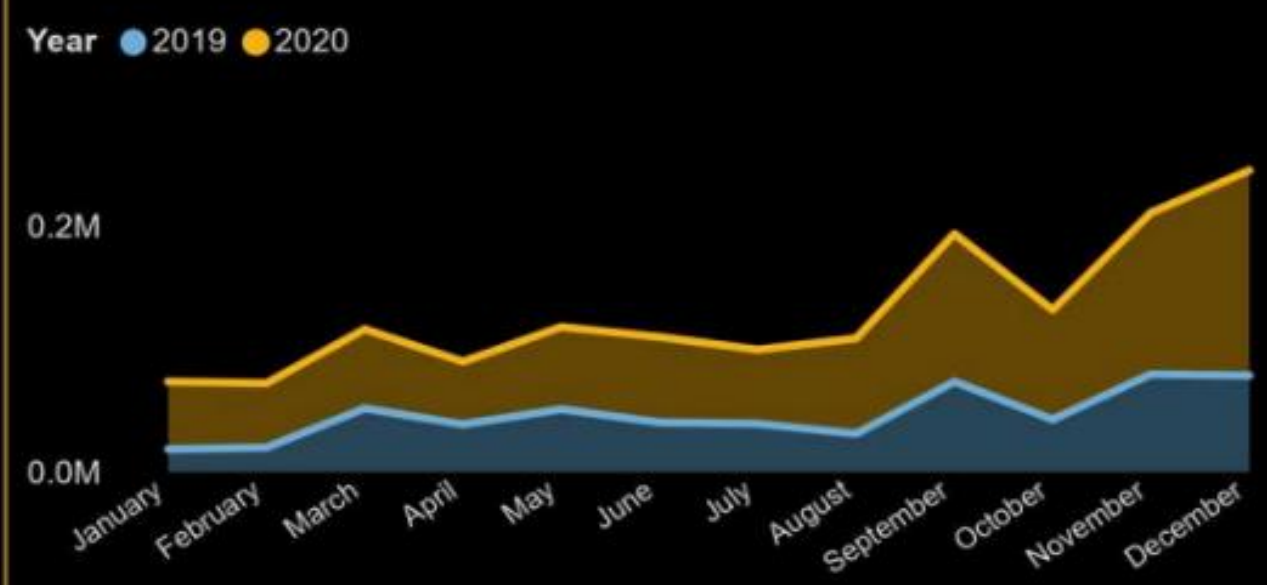
Sum of Sales by State



Sales by Payment Mode



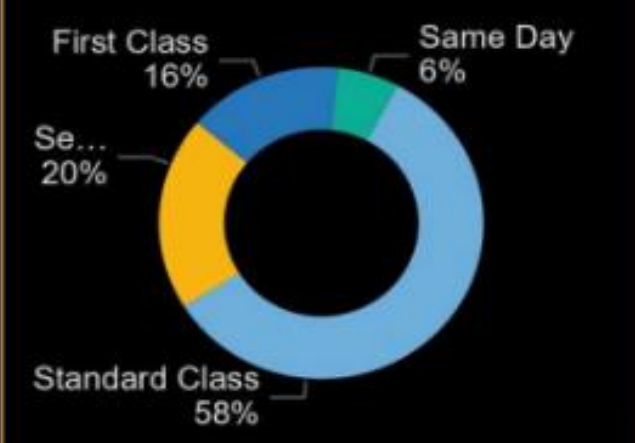
Seasonal Fluctuation of Sales Per Year



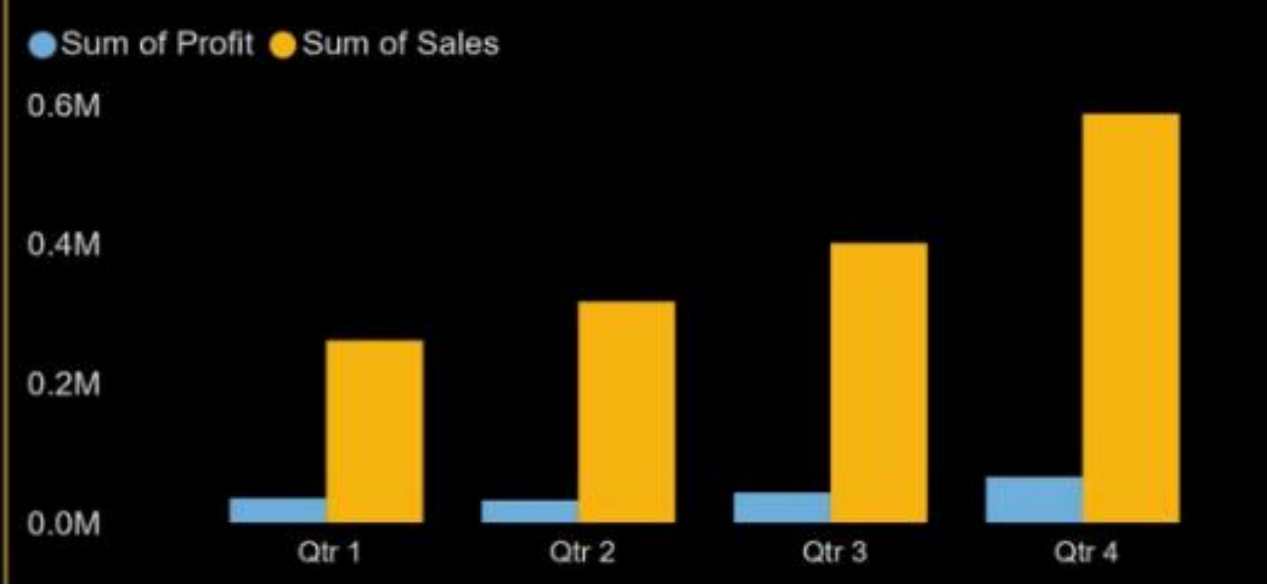
Sales by Category



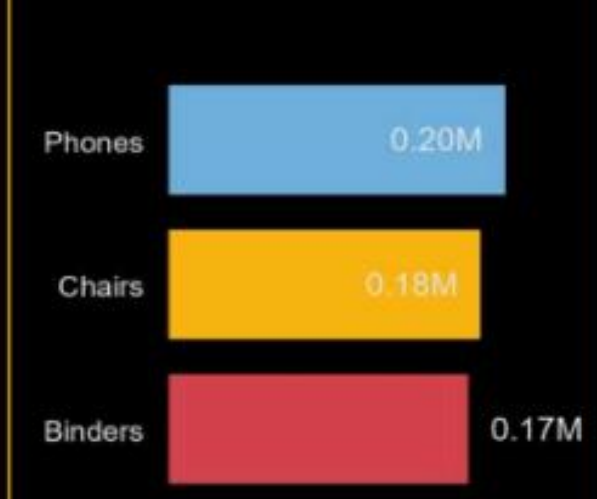
Sales by Ship Mode



Sales vs Profit Per Quarter



Top 3 Profitable Sub-Categories



Return Status



GENAI & LLM'S projects

Overview

I have hands-on **project experience in Generative AI**, working with **Deep Learning models, GANs, VAEs, and Large Language Models (LLMs)** to build real-world AI applications. These projects focus on solving practical problems in **computer vision, natural language processing, healthcare, and recommendation systems** using **Python, PyTorch, OpenAI APIs, and LangChain**.

Core Technical Experience

1. Generative AI & Deep Learning

- Designed and implemented **Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Generative Models**.
- Worked with **GANs** for image generation and **VAEs** for representation learning and recommendation systems.
- Applied **loss functions, optimizers, backpropagation, and evaluation metrics** for model performance analysis.

2. Computer Vision Projects

- Built **CNN-based models** for image classification and detection tasks using PyTorch and OpenCV.
- Implemented **Plant Disease Detection** using CNN architectures.
- Generated synthetic images using **GAN-based models** for data augmentation and visualization.

3. Natural Language Processing & LLM Projects

- Worked with **Large Language Models (LLMs)** to perform text classification, sentiment analysis, summarization, and text generation.
- Implemented **Medical Named Entity Recognition (NER)** using NLP techniques and OpenAI-based models.
- Developed **text-to-text and language translation** applications using transformer-based models.

4. LLM Applications & APIs

- Built **chatbot applications** using the **OpenAI API**.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines using **LangChain** to improve response accuracy.
- Integrated LLM workflows for question answering, summarization, and conversational AI use cases.

Implemented Projects

- **Breast Cancer Prediction** using ANN (PyTorch)
- **Plant Disease Detection** using CNN
- **Image Generation** using GANs
- **Text-to-Image and Image-to-Text Generation** using GAN models
- **Sentiment Analysis & Text Classification** using LLMs
- **Medical Named Entity Recognition (NER)** using NLP and OpenAI
- **Text Summarization** using Large Language Models
- **Text Generation** using GPT-2.5
- **Language Translation** using Transformer models
- **Fashion Recommendation System** using VAEs
- **Digit and Image Generation** using Generative AI models
- **Chatbot Development** using OpenAI API and LangChain

Tools & Technologies

Programming: Python

- **Frameworks & Libraries:** PyTorch, OpenCV, NLTK, Transformers
- **Generative Models:** GANs, VAEs, LLMs
- **Platforms & APIs:** OpenAI API, LangChain
- **Visualization:** Matplotlib, Seaborn

Appendix

[https://drive.google.com/drive/folders/1ODLJ6XBV5aRG8E0tO65yNVAye7lLewlK?usp=drive link](https://drive.google.com/drive/folders/1ODLJ6XBV5aRG8E0tO65yNVAye7lLewlK?usp=drive_link)

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